



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

FULL SYLLABUS

13-07-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

Test Type : SCORE-3.0

Test Pattern : JEE (Main)

TEST DATE : 13-07-2021

ANSWER KEY

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	D	B	D	A	A	B	A	A
SEC-1	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	B	A	B	A	D	A	A	B
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	0.50	10.00	320.00	8.00	2.00	30.00	0.12	2.00	3.00	3.00

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	D	B	B	B	D	C	D	B
SEC-1	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	D	A	D	D	C	B	A	B
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5.00	1.33	19.28 to 19.29	-2.27 to -2.25	4.00	2.40	88.50	9.74	2.00	5.00

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	D	C	B	C	B	B	C	B
SEC-1	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	B	B	B	C	C	C	A	D
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	14.00	1.00	0.22 or 0.23	9.00	1.41	49.30	90.00	9.00	0.00	-0.50

HINT – SHEET

PHYSICS

SEC-1

1. Ans (C)

$$\text{time of flight, } t = \frac{2v_0 \cos \theta}{g}$$

If projectile returns to point of projection,
displacement along x direction = 0

$$s_x = u_x t + \frac{1}{2} a_x t^2$$

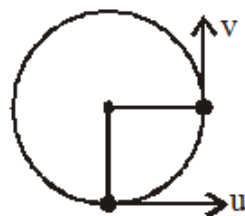
$$0 = (v_0 \sin \theta) \left(\frac{2v_0 \cos \theta}{g} \right) + \frac{1}{2} (-a) \left(\frac{2v_0 \cos \theta}{g} \right)^2$$

$$0 = v_0 \sin \theta - \frac{1}{2} \cdot a \left(\frac{2v_0 \cos \theta}{g} \right)$$

$$\tan \theta = \frac{a}{g} \text{ option (C)}$$

2. Ans (D)

By conservation of energy,



$$mgl = \frac{1}{2} mu^2 - \frac{1}{2} mv^2$$

$$v = \sqrt{u^2 - 2gl}$$

$$|\Delta \vec{v}| = \sqrt{u^2 + v^2}$$

$$= \sqrt{u^2 + u^2 - 2gl} = \sqrt{2(u^2 - gl)}$$

option(D)

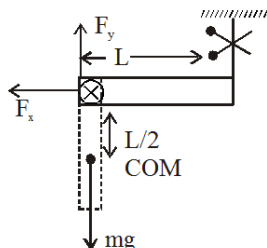
4. Ans (B)

$$M_1 a_1 = T, M_2 g - 2T = M_2 a_2, a_1 = 2a_2$$

$$\text{Hence } M_1(2a_2) = T \text{ and } M_2 g - 2M_1(2a_2) = M_2 a_2$$

$$M_2 g = a_2 (4M_1 + M_2) \text{ so } a_2 = \frac{M_2 g}{4M_1 + M_2}$$

5. Ans (D)



$$F_y - Mg = m\omega^2 R$$

$$F_y - Mg = m \frac{L}{2} \frac{3g}{L}$$

$$\therefore \omega = \sqrt{\frac{3g}{L}}$$

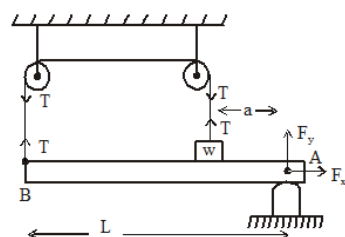
By energy conservation

$$F_y - mg = \frac{3mg}{2}$$

$$F_y = \frac{5mg}{2}$$

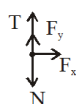
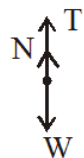
$$F_x = 0; F_y = \frac{5mg}{2}$$

6. Ans (A)



F.B.D of W

F.B.D for rod



$$F_y + T = N \dots\dots (i) \text{ (for rod)}$$

$$F_x = 0$$

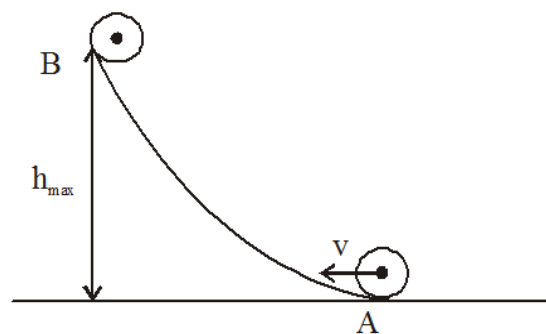
For equilibrium $\tau = 0$

$$\therefore \tau \text{ about B, } N(L - a) = F_y L \dots\dots (ii)$$

solving (i) & (ii) we get

$$F_y = \frac{W(L - a)}{(L + a)}$$

7. Ans (A)



Applying energy conservation at A & B

$$\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = mgh \left\{ \begin{array}{l} \because I = \frac{mR^2}{2} \\ R\omega = v \end{array} \right\}$$

$$\therefore h = \frac{3v^2}{4g}$$

8. Ans (B)

The wave propagate in x-direction while electric field oscillates in y-direction. Thus the electric field resides in the x-y plane. Since the magnetic field is perpendicular to both E & propagation direction, it must be in x-z plane $\therefore$

$$B_x = 0, B_y = 0, B_z \neq 0 \text{ and } B_z = \frac{E_y}{C} \Rightarrow B_z = \frac{0.5 \cos[2\pi \times 10^8 (t - \frac{x}{c})]}{3 \times 10^8}$$

9. Ans (A)

The given equation can be represented as

$$y = 2 \times 10 \cos\left(\frac{2\pi x}{50}\right) \sin\left(\frac{2\pi}{50} 5000t\right)$$

comparing it with standard equation of stationary wave

$$\frac{2\pi}{\lambda} = \frac{2\pi}{50}$$

$$\lambda = 50 \text{ units}$$

$$\text{so loop length } \lambda/2 = 25 \text{ units.}$$

10. Ans (A)

The given lens is a convex lens. Let the magnification be m, then for real image

$$\frac{1}{mx} + \frac{1}{x} = \frac{1}{f} \dots\dots(i)$$

$$\text{and for virtual image } \frac{1}{-my} + \frac{1}{y} = \frac{1}{f} \dots\dots(ii)$$

From Eq. (i) and Eq.(ii), we get

$$f = \frac{x+y}{2}$$

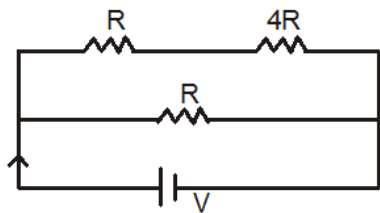
12. **Ans (B)**

$$v_e = \sqrt{2}v_0 = 1.414 v_0 \% \text{ increase in orbital velocity} = \frac{v_0 - v_0}{v_0} \times 100 = 41.4\%$$

13. **Ans (B)**

$$\begin{aligned}\vec{\tau} &= \vec{P} \times \vec{E} \\ &= 9 \times 10^{-6} (\hat{i} - \hat{j} + \hat{k}) \times 0.20 \times 100 \hat{i} \\ \vec{\tau} &= 18 \times 10^{-7} (\hat{k} - \hat{j}) \\ |\vec{\tau}| &= 25.45 \times 10^{-7} \text{ N-m}\end{aligned}$$

14. **Ans (A)**



$$\begin{aligned}W_1 &= \frac{V^2}{R} = \frac{V^2}{\frac{V^2}{P}} = P \\ W_2 &= \left(\frac{V}{5R}\right)^2 \cdot 4R = \frac{4V^2 \cdot P}{25 V^2} = \frac{4P}{25}\end{aligned}$$

15. **Ans (B)**

$$\begin{aligned}\phi &= \frac{\mu_0 i}{2R} \cdot \pi r \\ \text{compare with } \phi &= \mu_i \\ \Rightarrow \left[M = \frac{\mu_0 \pi r^2}{2R} \right]\end{aligned}$$

16. **Ans (A)**

$$\begin{aligned}V_s &= \tan \theta \cdot v + C \\ 4V &= \tan \theta \cdot 2v_0 + C \\ 10V &= \tan \theta \cdot 4v_0 + C \\ 6V &= 2v_0 \tan \theta \Rightarrow \tan \theta = \frac{3V}{v_0} \\ 4V &= 6V - \frac{W_e}{e} \\ W_e &= (2V)e \\ h &= \frac{(3V)e}{v_0}\end{aligned}$$

17. **Ans (D)**

$$\begin{aligned}\lambda_\alpha &= \frac{1}{3240} \text{ per year and } \lambda_\beta = \frac{1}{810} \text{ per year and} \\ \text{it is given that the fraction of the remained} \\ \text{activity } \frac{A}{A_0} &= \frac{1}{4} \\ \text{Total decay constant} \\ \lambda &= \lambda_\alpha + \lambda_\beta = \frac{1}{3240} + \frac{1}{810} = \frac{1+4}{3240} \text{ per year} \\ &= \frac{1}{648} \text{ per year}\end{aligned}$$

$$\text{We know that } A = A_0 e^{-\lambda t} \Rightarrow t = \frac{1}{\lambda} \log_e \frac{A_0}{A}$$

$$\begin{aligned}\Rightarrow t &= \frac{1}{\lambda} \log_e 4 = \frac{2}{\lambda} \log_e 2 \\ t &= 648 \times 2 \times 0.693 = 898 \text{ years}\end{aligned}$$

18. **Ans (A)**

After falling first fragment on the ground, horizontal velocity of c.m. remains unchanged but vertical acceleration decreases.

19. **Ans (A)**

$$\begin{aligned}e &= \frac{\left| \frac{\vec{v}_2 \cdot \Delta \vec{p}}{|\Delta \vec{p}|} \right|}{\left| \frac{\vec{v}_1 \cdot \Delta \vec{p}}{|\Delta \vec{p}|} \right|} = \frac{|\vec{v}_2 \cdot (\vec{v}_2 - \vec{v}_1)|}{|\vec{v}_1 \cdot (\vec{v}_2 - \vec{v}_1)|} \\ &= \frac{|(\hat{i} + 2\hat{j}) \cdot (-\hat{i} - \hat{j})|}{|(2\hat{i} + 3\hat{j}) \cdot (-\hat{i} - \hat{j})|} = \frac{3}{5}\end{aligned}$$

20. **Ans (B)**

More slit width more will be light intensity coming from it and width of central maxima decreases with slit width, so A will move towards O.

PHYSICS

SEC-2

1. **Ans (0.50)**

Tension of rope is maximum at lowest point

$$T_{\max} - mg = \frac{mv^2}{\ell} \dots (1)$$

By energy conservation,

$$mg \frac{\ell}{2} = \frac{1}{2} mv^2 \Rightarrow v = \sqrt{g\ell}$$

from (1),

$$T_{\max} - mg = \frac{m(g\ell)}{\ell} \Rightarrow T_{\max} = 2mg$$

for 8 kg block, $T_{\max} = f_L$

$$2mg = \mu (8g)$$

$$\mu = 0.5 \text{ option(B)}$$

2. **Ans (10.00)**

$$a_t = \frac{dv}{dt} = \frac{t}{5} \text{ m/s}^2$$

$$a_r = \frac{v^2}{R} = \frac{t^4}{5000}$$

$$a_t = a_r \Rightarrow \frac{t}{5} = \frac{t^4}{5000} \Rightarrow t^3 = 1000$$

$$t = 10 \text{ sec.}$$

3. **Ans (320.00)**

When the two positions of the resonance are obtained at distances ℓ_1 & ℓ_2 respectively then the velocity V of sound is given by $V = 2f(\ell_2 - \ell_1)$

$$V = 2 \times 1600 \times 10 \times 10^{-2} = 320 \text{ m/s}$$

4. **Ans (8.00)**

Frequency observed for A

$$(f_A) = \left(\frac{340 - 20}{340 - 20 + 20} \right) 68$$

$$= \frac{32}{34} \times 68 = 64$$

Frequency observed for B

$$(f_B) = \left(\frac{340 + 20}{340 + 20 - 20} \right) 68$$

$$= \left(\frac{36}{34} \right) 68 = 72$$

$$\text{So beat frequency } f_B - f_A = 72 - 64 = 8$$

5. **Ans (2.00)**

w.r.t stationary block

$$T = 2\pi\sqrt{\frac{\mu}{k}} \text{ (for oscillation)}$$

so time of contact $T/2$

$$t = \frac{T}{2} = \pi\sqrt{\frac{\frac{4 \times 4}{4+4}}{\pi^2}} = \sqrt{2} \text{ sec.}$$

7. **Ans (0.12)**

$\tau = R_{Th} \cdot C$, R_{Th} is the thevenin resistance at the capacitor terminals.

$$R_{Th} = 8 + (20 \parallel (9 + (70 \parallel 30))) = 20 \text{ K}\Omega$$

$$\tau = 0.12 \text{ s}$$

8. **Ans (2.00)**

$$P = Li \frac{di}{dt} = 8i \therefore di = dt$$

$$\Rightarrow \int_{i_2}^{i_4} di = \int_2^4 dt \quad i_4 - i_2 = 2A$$

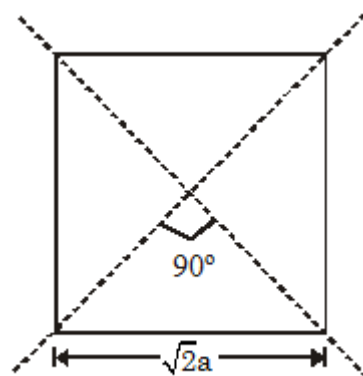
9. **Ans (3.00)**

White spot on screen would be central maxima where

$$\Delta x = 0 \quad y = \frac{d}{2} - \frac{d}{8} = \frac{3d}{8}$$

10. **Ans (3.00)**

Consider prism of mass $4m$ by joining 4 prism given in question. total MOI of this system will be



$$I = (4I_{\text{prism}}) = \frac{(4m)(\sqrt{2}a)^2}{6}$$

$$\Rightarrow I_{\text{prism}} = \frac{ma^2}{3}$$

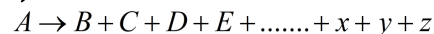
CHEMISTRY

SEC-1

1. **Ans (B)**

Colloidal particles in As_2S_3 solution are negatively charged

2. **Ans (C)**



$$t=0 \quad P_0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0+0+0$$

$$t=t \quad P_0 - x \quad x \quad x \quad x \quad x \quad x \quad x \quad x$$

$$t=\infty \quad 0 \quad P_0 \quad P_0 \quad P_0 \quad P_0 \quad P_0 \quad P_0 \quad P_0$$

$$\therefore \text{At } t=t; P_0 + 24x = p$$

$$x = \frac{P - P_0}{24}$$

$$\& P_0 - x = P_0 - \frac{P}{24} + \frac{P_0}{24} = \frac{25P_0 - P}{24}$$

$$K = \frac{1}{t} \ln \left(\frac{24P_0}{25P_0 - P} \right) = \frac{2.303}{t} \log \left(\frac{24P_0}{25P_0 - P} \right)$$

3. **Ans (D)**

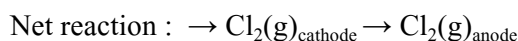
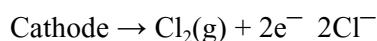
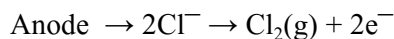
$$n = 4, m = -3$$

i.e. only possible value of l is 3.

i.e. Orbital angular momentum

$$= \sqrt{l(l+1)} \frac{h}{2\pi} = \frac{2\sqrt{3}h}{2\pi} = \frac{\sqrt{3}h}{\pi}$$

4. **Ans (B)**



Applying nerst equation

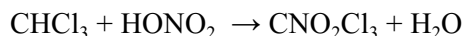
$$E_{\text{cell}} = 0 - \frac{0.059}{2} \log \frac{P_1}{P_2} = \frac{0.059}{2} \log \frac{P_2}{P_1}$$

If $P_2 > P_1$ then cell will be spontaneous

Since E_{cell} will be positive.

5. **Ans (B)**

When chloroform is treated with concentrated nitric acid, its hydrogen is replaced by nitrogroup



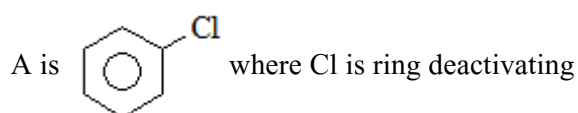
6. **Ans (B)**

Hydrazine does not have carbon atoms so it does not form NaCN and hence does not give a positive Lassaigne's test

7. **Ans (D)**

Option (D) have α -H atom $\therefore$ it show tautomerism

8. **Ans (C)**



& less reactive.

9. **Ans (D)**

Sunlight / $h\nu$ is responsible for formation of F. R.

10. **Ans (B)**

Only terminal alkynes can react.

11. **Ans (C)**

Only 1° amide can undergo Hoffmann Bromamide Reaction.

12. **Ans (A)**

According to IUPAC ligands are named in alphabetical order

13. **Ans (D)**

Being the first alkali metal Li shows some difference with remaining alkali metal.

14. **Ans (A)**

NaCN acts as depressant for ZnS for the mixture of PbS and ZnS ores.

15. **Ans (D)**

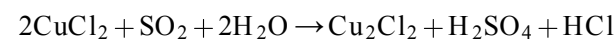
Electronegativity of elements on Pauling's scale :

$$B = 2, Al = 1.5, Ga = 1.6, In = 1.7, Tl = 1.8$$

16. **Ans (D)**

$x = 101^\circ$, $y = 79^\circ$ & $z = 118^\circ$ According to NCERT.

17. **Ans (C)**



$x = \text{SO}_2$ gas.

CHEMISTRY

SEC-2

1. **Ans (5.00)**

Let x mL of 0.2 M H_2SO_4 is required

	$2\text{NaCN} + \text{H}_2\text{SO}_4 \longrightarrow 2\text{HCN} + \text{Na}_2\text{SO}_4$	
Initial	50 mL, 0.2 M x mL, 0.2 M	— —
final	$(10 - 0.4x)$ mmol	— $0.4x$ mmol

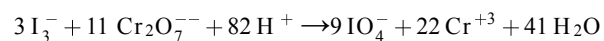
$$\text{pH} = \text{pK}_a + \log \frac{[\text{NaCN}]}{[\text{HCN}]}$$

$$10 = 9.4 + \log \frac{(10 - 0.4x)}{0.4x}$$

on solving $x = 5$ mL

2. Ans (1.33)

On balancing equation



$$(a+b+c)/(d+e+f) \Rightarrow \frac{(3+11+82)}{(9+22+41)}$$

$$= \frac{4}{3} = 1.33$$

3. Ans (19.28 to 19.29)

$$C_{v(\text{mix})} = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2}$$

$$= \frac{3 \times \frac{5R}{2} + \frac{2 \times 5R}{2}}{5} = \frac{5R}{2}$$

$$\Delta U = nC_v (T_2 - T_1)$$

$$\Delta U = 5 \times \frac{5}{2} \times 2 [T_2 - 300] \quad \dots (1)$$

$$-P_{\text{ext}} \left[\frac{T_2}{P_2} - \frac{T_1}{P_1} \right] nR = nC_v (T_2 - T_1)$$

$$-10 \left[\frac{T_2}{10} - \frac{300}{1} \right] nR = n \times \frac{5R}{2} [T_2 - 300]$$

$$T_2 = \frac{1500}{1.4} \text{ K}$$

$$\Delta U = 5 \times \frac{5}{2} \left[\frac{1500}{1.4} - 300 \right]$$

$$= 25 \left(\frac{1080}{1.4} \right) \times \frac{1}{1000} \text{ kcal}$$

$$= \frac{27}{1.4} \text{ kcal}$$

$$= 19.2857$$

$$\simeq 19.29$$

4. Ans (-2.27 to -2.25)

$$dG = VdP - SdT$$

$$T = \text{const} \quad dT = 0$$

$$\int dG = nRT \int \frac{1}{P} \cdot dP$$

$$\Delta G = 2 \times 2 \times 353 \ln \frac{0.2}{1} = -2.27 \text{ kcal}$$

5. Ans (4.00)

$$\Lambda_m = \Lambda_m^\infty$$

$$\frac{k \times 1000}{M} = 63 + 67$$

$$\frac{2.6 \times 10^{-6} \times 1000}{M} = 130$$

$$M = 2 \times 10^{-5} = \text{solubility in mol/L}$$

$$K_{sp}(\text{AgCl}) = (2 \times 10^{-5})^2 = 4 \times 10^{-10}$$

$$\text{Hence } x = 4$$

6. Ans (2.40)

When AgCl Starts precipitating

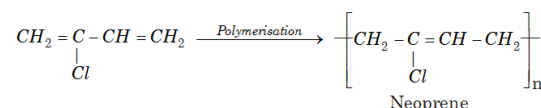
$$[\text{Ag}^+] = \frac{10^{-10}}{0.06}$$

at that time conc of $[\text{I}^-]$

$$= \frac{4 \times 10^{-16}}{\frac{10^{-10}}{0.06}} = 2.4 \times 10^{-7}$$

7. Ans (88.50)

$\text{CH}_2 = \underset{\text{Cl}}{\text{C}} - \text{CH} = \text{CH}_2$ is the monomer of neoprene.



Molecular mass

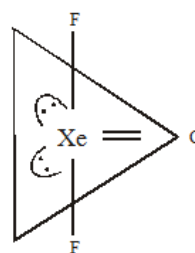
$$= (12 \times 4) + (1 \times 5) + (1 \times 35.5)$$

$$= 48 + 5 + 35.5$$

$$= 88.5$$

10. Ans (5.00)

XeOF_2 is 'T' shaped



with Xe in sp^3d hybridization

The geometry is trigonal bipyramidal

No. of orbitals involved in hybridisation = 5

MATHEMATICS

SEC-1

1. Ans (B)

$$\text{Let } \vec{s} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{p} \cdot \vec{s} = 0 \text{ gives } x - y = 0 \quad \dots(1)$$

$$\text{and } [\vec{q} \ \vec{r} \ \vec{s}] = 0 \text{ gives } x + y + z = 0 \quad \dots(2)$$

$$\text{i.e. } x = y \text{ and } z = -2x \text{ and } x^2 + y^2 + z^2 = 1 \dots$$

(3)

$$\text{i.e. } x = \pm \frac{1}{\sqrt{6}}, y = \pm \frac{1}{\sqrt{6}}, z = \mp \frac{2}{\sqrt{6}}$$

$$\vec{s} = \pm \left(\frac{\hat{i} + \hat{j} - 2\hat{k}}{\sqrt{6}} \right)$$

2. Ans (C)

p	q	$\sim p \vee q$	$\sim p \wedge \sim q$	$(\sim p \vee q) \wedge (\sim p \wedge \sim q)$
T	T	T	F	F
T	F	F	F	F
F	T	T	F	F
F	F	T	T	T

3. Ans (D)

It's D.R is $\{3, 1, 0\}$

i.e. option (1) & (2) are correct

Any point onto line is $3 + 3\lambda, 2 + \lambda, 1$

hence for $\lambda = 1$, option (3) is correct

Line & plane are parallel to each other hence option (4) is incorrect.

4. Ans (C)

P	$\sim P$	$P \Rightarrow \sim P$	$\sim P \Rightarrow P$	$(P \Rightarrow \sim P) \wedge (\sim P \Rightarrow P)$
T	F	F	T	F
F	T	T	F	F

5. Ans (B)

Set $S \times S$ will have $4 \times 4 = 16$ elements.

Number of elements except (1, 2), (2, 3) and (3, 4) in $S \times S = 16 - 3 = 13$

Thus number of subsets of $S \times S$ containing (1, 2), (2, 3) and (3, 4) $= 2^{13} = 8192$.

6. Ans (C)

$$\sim \{p \wedge (q \rightarrow r)\} \equiv \sim p \vee \sim (q \rightarrow r)$$

$$[\because \sim (a \wedge b) \equiv \sim a \vee \sim b]$$

$$\equiv \sim p \vee (q \wedge \sim r) \quad [\because \sim (a \rightarrow b) \equiv a \wedge \sim b]$$

7. Ans (B)

$$\text{Let } g(x) = f(x) - x^3$$

$\Rightarrow g(x)$ has at least 4 real roots which are $x = 1, 2, 3, 4$

$\Rightarrow g'(x) = 0$ has atleast 3 real roots in (1, 4)

$\Rightarrow g''(x) = 0$ has atleast 2 real roots in (1, 4)

$\Rightarrow g'''(x) = f'''(x) - 6 = 0$ has atleast 1 real root in (1, 4)

8. Ans (B)

$$ydx - xdy + 3x^2y^2e^{x^3}dx = 0$$

$$\Rightarrow \frac{ydx - xdy}{y^2} + 3x^2e^{x^3}dx = 0$$

$$\Rightarrow d\left(\frac{x}{y}\right) + d\left(e^{x^3}\right) = 0 \Rightarrow \frac{x}{y} + e^{x^3} = C$$

9. Ans (C)

We know that

$$\tan \alpha = \cot \alpha - 2\cot(2\alpha) \dots \dots \dots (i)$$

Putting

$$\alpha = \theta, 2\theta, 2^2\theta \dots \dots \dots, 2^{10}\theta \text{ in (i),}$$

we get

$$\tan \theta = (\cot \theta - 2 \cot (2\theta))$$

$$2 (\tan(2\theta)) = 2 (\cot(2\theta) - 2 \cot (2^2 \theta))$$

$$2^2 (\tan(2^2\theta)) = 2^2 (\cot(2^2\theta) - 2 \cot (2^3 \theta))$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$2^{10} (\tan(2^{10}\theta)) = 2^{10} (\cot(2^{10}\theta) - 2 \cot (2^{11} \theta))$$

adding

$$\tan \theta + 2 \tan(2\theta) + 2^2 \tan(2^2\theta) + \dots \dots \dots +$$

$$2^{10} \tan(2^{10}\theta) = \cot \theta - 2^{11} \cot(2^{11}\theta)$$

10. Ans (B)

Locus will be : $3|x| = 2|y|$

11. **Ans (B)**

$$\text{Let } S : x^2 + y^2 + 2gx + 2fy + c = 0$$

$$S_1 : x^2 + y^2 - 9 = 0$$

$$S_2 : x^2 + y^2 - 2x - 4y + 2 = 0$$

Common chord of S and S_1 passing through (0, 0)

$$S - S_1 = 0, 2gx + 2fy + c + 9 = 0$$

$$c = -9$$

common chord of S and S_2 passing through (1, 2)

$$S - S_2 = 0,$$

$$2gx + 2fy + 2x + 4y + C - 2 = 0$$

$$2g + 4f + 2 + 8 - 9 - 2 = 0$$

$$2g + 4f - 1 = 0$$

$$\text{So locus is } -2x - 4y - 1 = 0$$

$$\Rightarrow 2x + 4y + 1 = 0$$

12. **Ans (C)**

Point lie on directrix so angle will be 90° .

13. **Ans (B)**

Point should lie on director circle $x^2 + y^2 = 41$

14. **Ans (B)**

$$\det A = \begin{vmatrix} 7 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 3 & 1 \end{vmatrix} = 7 + 1 \times (-6 + 1) = 2$$

$$|\text{adj}(\text{adj}(\text{adj}(\text{adj } A)))| = |A|^{2^4} = 2^{16}$$

15. **Ans (B)**

$$\left. \begin{array}{l} ar^3 = 7! \\ ar^6 = 8! \end{array} \right\} \Rightarrow r = 2, a = \frac{7!}{8} = 630$$

$$\text{Now } a \left(\frac{r^n - 1}{r - 1} \right) = S$$

$$\Rightarrow 630 \left(\frac{2^n - 1}{2 - 1} \right) = 19530$$

$$\Rightarrow 2^n - 1 = 31$$

$$\Rightarrow 2^n = 32$$

$$\Rightarrow n = 5$$

16. **Ans (C)**

$$\lim_{x \rightarrow 0} \left[\frac{\tan 3x}{x} \right] = 3, \lim_{x \rightarrow 0} \left[\frac{\sin x}{x} \right] = 1$$

$$\text{curve is } 5 + x^2 \sqrt{y - 2} = y^2 - 3x$$

$$\left. \frac{dy}{dx} \right|_{(1, 3)} = \frac{10}{11}$$

$$\text{slope of normal is } \frac{-11}{10}$$

$$\text{equation of normal is } 11x + 10y = 41$$

17. **Ans (C)**

$$f(x) \text{ is continuous at } x^3 - 5x = 2x^2 - 6$$

$$x^3 - 2x^2 - 5x + 6 = 0$$

$$x = -2, 1, 3$$

$$x_1 = -2, x_2 = 1, x_3 = 3$$

18. **Ans (C)**

$$I = \int \frac{\left(\frac{1}{x} - \frac{1}{x^3} \right) dx}{\sqrt{1 + \frac{3}{x^2} + \frac{1}{x^4}}} = \int \frac{\left(1 - \frac{1}{x^2} \right) dx}{\sqrt{x^2 + 3 + \frac{1}{x^2}}}$$

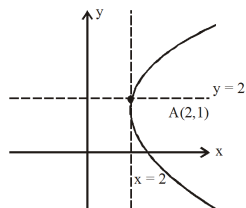
$$= \int \frac{\left(1 - \frac{1}{x^2} \right) dx}{\sqrt{\left(x + \frac{1}{x} \right)^2 + 1}}$$

$$\text{Now put } x + \frac{1}{x} = t$$

$$\left(1 - \frac{1}{x^2} \right) dx = dt = \int \frac{dt}{\sqrt{t^2 + 1}}$$

$$= \ln \left(x + \frac{1}{x} + \sqrt{x^2 + \frac{1}{x^2} + 3} \right) + c$$

19. Ans (A)



$$x = y^2 - 2y + 3$$

$$x = (y - 1)^2 + 2$$

$$x - 2 = (y - 1)^2$$

$$\Rightarrow (y - 1)^2 = 1(x - 2)$$

Now let slope of tangent is an

$$\text{so } m = -1/2$$

$\therefore$ equation of tangent will be

$$y - 1 = m(x - 2) + \frac{1}{4m}$$

$$y - 1 = -\frac{1}{2}(x - 2) + \frac{1}{4(-1/2)}$$

$$y - 1 = -\frac{1}{2}(x - 2) - \frac{1}{2}$$

$$2y - 2 = -x + 2 - 1$$

$$2y - 2 = -x + 1$$

$$x + 2y = 3$$

20. Ans (D)

$$p = 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots - \infty$$

$$p = \frac{1}{1 - 1/2} = 2$$

Let n^{th} term of 1, 1, 2, 3, 4, ---- .

$= (n - 1)^{\text{th}}$ term of 1, 2, 3, 4, -----

$$= 1 + (n - 2) \times 1 = n - 1$$

now $q = 20^{\text{th}}$ term of 1, 1, 2, 3, 4, -----

$$= 20 - 1$$

$$= 19$$

so desired 2 quadratic equation will be

$$x^2 - (p + q)x + pq = 0$$

$$x^2 - 21x + 38 = 0$$

MATHEMATICS

SEC-2

1. Ans (14.00)

$$\frac{1.5 + 2.4 + 3.0 + 4.f}{5 + 4 + 0 + f} = 3 \Rightarrow f = 14$$

2. Ans (1.00)

$$\frac{3 \cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = \frac{\cos^2 2x + 3}{1 - \frac{\sin^2 2x}{4}} = 4$$

$$x^2 + 3 = 4 \Rightarrow x = \pm 1$$

3. Ans (0.22 or 0.23)

Let E_1, E_2 & E_3 be the event that letter has come from RAJASTHAN, MAHARASTRA and MADRAS resp.

A be the event that consecutive letters RA are visible.

$$\therefore P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$$

$$P\left(\frac{A}{E_1}\right) = \frac{1}{8}, P\left(\frac{A}{E_2}\right) = \frac{2}{9}, P\left(\frac{A}{E_3}\right) = \frac{1}{5}$$

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1) P\left(\frac{A}{E_1}\right)}{P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) + P(E_3)P\left(\frac{A}{E_3}\right)}$$

$$= \frac{\frac{1}{3} \times \frac{1}{8}}{\frac{1}{3} \times \frac{1}{8} + \frac{1}{3} \times \frac{2}{9} + \frac{1}{3} \times \frac{1}{5}} = \frac{45}{197}$$

4. Ans (9.00)

$$x^3 - x + 3 = 0$$

$$\alpha + \beta + \gamma = 0, \alpha\beta\gamma = -3$$

$$\text{Now } (\alpha + \beta)^3 + (\beta + \gamma)^3 + (\gamma + \alpha)^3$$

$$= (-\gamma)^3 + (-\alpha)^3 + (-\beta)^3$$

$$= -[\alpha^3 + \beta^3 + \gamma^3]$$

$$= -3\alpha\beta\gamma, (\because \alpha + \beta + \gamma = 0)$$

$$= -3(-3) = 9$$

5. Ans (1.41)

$$e = \sqrt{1 + \frac{\sin^2 2\theta}{\cos^2 \theta}} = \sqrt{1 + \frac{4\sin^2 \theta \cos^2 \theta}{\cos^2 \theta}}$$

$$\sqrt{1 + 4\sin^2 \theta}$$

$$e_{\max} = \sqrt{1 + 4\sin^2 \left(\frac{\pi}{6}\right)} = \sqrt{2}$$

6. Ans (49.30)

Alphabetical order of letters are

A, E, H, J, R, S

A						5!
E						5!
H						5!
J						5!
R	A	E				3!
R	A	H				3!
R	A	J	E	H	S	1
R	A	J	E	S	H	

Total number of words before the word

RAJESH is 493

7. Ans (90.00)

$$(1+x)^{89} = {}^{89}C_0 x^0 + {}^{89}C_1 x^1 + \dots + {}^{89}C_{89} x^{89}$$

integrating from 0 to 1

$$\left[\frac{(1+x)^{90}}{90} \right]_0^1 = \left[\frac{{}^{89}C_0 x^1}{1} + \frac{{}^{89}C_1 x^2}{2} + \dots + \frac{{}^{89}C_{89} x^{90}}{90} \right]_0^1$$

$$\frac{2^{90} - 1}{90} = \frac{{}^{89}C_0}{1} + \frac{{}^{89}C_1}{2} + \dots + \frac{{}^{89}C_{89}}{90} \dots (i)$$

similarly integrating from

(1 to ω) and (1 to ω^2)

we have (where ω is complex cube root of unity)

$$0 = \frac{{}^{89}C_0 \omega^1}{1} + \frac{{}^{89}C_1 \omega^2}{2} + \dots + \frac{{}^{89}C_{89} \omega^{90}}{90} \dots (ii)$$

$$0 = \frac{{}^{89}C_0 \omega^2}{1} + \frac{{}^{89}C_1 \omega^4}{2} + \dots + \frac{{}^{89}C_{89} \omega^{180}}{90} \dots (iii)$$

Now (1) + ω^2 (2) + ω (3)

we get the required result

8. Ans (9.00)

$$\lim_{n \rightarrow \infty} \frac{1}{\left(\frac{x-3}{5}\right)^n + 5 - \frac{1}{n}} = \frac{1}{5}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{\left(\frac{x-3}{5}\right)^n + 5} = \frac{1}{5} \left(\because \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left(\frac{x-3}{5}\right)^n = 0$$

$$\Rightarrow -1 < \frac{x-3}{5} < 1 \Rightarrow -2 < x < 8$$

9. Ans (0.00)

$$f(x) = x^n \Rightarrow f(1) = 1$$

$$f'(x) = nx^{n-1} \Rightarrow f'(1) = n$$

$$f''(x) = n(n-1)x^{n-2} \Rightarrow f''(1) = n(n-1)$$

$$\Rightarrow f(1) - \frac{f'(1)}{1} + \frac{f''(1)}{2} - \dots + \frac{(-1)^n f^{(n)}(1)}{n!}$$

$$= {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots + (-1)^n {}^nC_n = 0$$

10. **Ans (-0.50)**

$$\begin{aligned}
 I &= \int_{-1/2}^{1/2} \left([x] + \ln \left(\frac{1-x}{1+x} \right) \right) dx \\
 &= \int_{-1/2}^{1/2} [x] dx + \int_{-1/2}^{1/2} \ln \left(\frac{1-x}{1+x} \right) dx \\
 &= \int_{-1/2}^0 [x] dx + \int_0^{1/2} [x] dx + 0 \\
 &\text{(because } \ln \left(\frac{1-x}{1+x} \right) \text{ is odd function)} \\
 &= \int_{-1/2}^0 (-1) dx + 0 + 0 \\
 &= (-x)_{-1/2}^0 = -(0 + 1/2) = -1/2
 \end{aligned}$$